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|-------------------------------------------------------|-------------------------------------------|
| Program: <b>Diploma in Food Processing Technology</b> |                                           |
| Course Code: <b>4421</b>                              | Course Title: <b>Food Microbiology II</b> |
| Semester: <b>4</b>                                    | Credits: <b>3</b>                         |
| Course Category:                                      |                                           |
| Periods per week: <b>4 (L:3 T:1 P: 0)</b>             | Periods per semester: <b>60</b>           |

### Course Objectives:

1. To understand various quantitative and qualitative enumeration methods to detect microbes and microbial toxins in food
2. To understand the various enzymes and microorganisms used in processing of food
3. To impart knowledge on various fermented food products

### Course Prerequisites:

| Topic                              | Program/Course Name  |
|------------------------------------|----------------------|
| Microorganism associated with food | Food Microbiology II |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                        | Duration (Hours) | Cognitive Level |
|-----------------|------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Illustrate the qualitative methods to detect microbes and microbial toxins in food | 14               | Understanding   |
| CO2             | Summarize the quantitative methods to detect microbes and microbial toxins in food | 16               | Understanding   |
| CO3             | Identify the beneficial organisms in the food processing industry                  | 15               | Understanding   |
| CO4             | Describe various fermented foods                                                   | 15               | Understanding   |

## CO–PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     |     |     |     |     |
| CO2             | 2   |     |     |     |     |     |     |
| CO3             | 2   |     |     |     |     |     |     |
| CO4             | 2   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline:

| Module Outcomes                                                                                                                                                                                           | Description                                                                                        | Duration (Hours) | Cognitive Level |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>                                                                                                                                                                                                | <b>Illustrate the qualitative methods to detect microbes and microbial toxins in food</b>          |                  |                 |
| M1.01                                                                                                                                                                                                     | Explain ELISA technique                                                                            | 5                | Understanding   |
| M1.02                                                                                                                                                                                                     | Discuss RIA technique                                                                              | 5                | Understanding   |
| M1.03                                                                                                                                                                                                     | Outline Immunomagnetic methods                                                                     | 4                | Understanding   |
| <b>Contents:</b> Concepts and principles of Immunological methods of qualitative detection of microbes and microbial toxins in food-ELISA, RIA, Immunomagnetic methods                                    |                                                                                                    |                  |                 |
| <b>CO2</b>                                                                                                                                                                                                | <b>Summarize the quantitative methods to detect microbes and microbial toxins in food</b>          |                  |                 |
| M2.01                                                                                                                                                                                                     | Identify various physical Quantitative methods to detect microbes and microbial toxins in food     | 5                | Understanding   |
| M2.02                                                                                                                                                                                                     | Identify the various chemical Quantitative methods to detect microbes and microbial toxins in food | 5                | Understanding   |
| M2.03                                                                                                                                                                                                     | Summarize different tests for bacterial toxins in foods                                            | 5                | Understanding   |
|                                                                                                                                                                                                           | Series Test –I                                                                                     | 1                |                 |
| <b>Contents:</b> Quantitative methods for microbial enumeration in foods, Physical (Impedance, Microcalorimetry, Flow cytometry) , Chemical (LAL, ATPase, Radiometry), Test for bacterial toxins in foods |                                                                                                    |                  |                 |

|                                                                                                                                             |                                                                          |   |               |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---|---------------|
| <b>CO3</b>                                                                                                                                  | <b>Identify the beneficial organisms in the food processing industry</b> |   |               |
| M3.01                                                                                                                                       | Summarize biotics and probiotics in food                                 | 5 | Understanding |
| M3.02                                                                                                                                       | Outline the beneficial microorganisms in food processing                 | 5 | Understanding |
| M3.03                                                                                                                                       | State different microbial enzymes used in Food Processing                | 5 | Understanding |
| <b>Contents:</b> Introduction to biotics and probiotics, Microbial enzymes in Food Processing, Beneficial microorganisms in food processing |                                                                          |   |               |
| <b>CO4</b>                                                                                                                                  | <b>Describe various fermented foods</b>                                  |   |               |
| M4.01                                                                                                                                       | Describe various Fermented milk and milk products                        | 5 | Understanding |
| M4.02                                                                                                                                       | Identify different fermented fruits and vegetables                       | 5 | Understanding |
| M4.03                                                                                                                                       | Outline various fermented fish and Fermented meats                       | 5 | Understanding |
|                                                                                                                                             | Series Test–II                                                           | 1 |               |
| <b>contents:</b> Fermented milk and milk products, fermented fruits and vegetables, Fermented fish, Fermented meats                         |                                                                          |   |               |

### Text/Reference

| <b>T/R</b> | <b>Book Title/Author</b>                                                           |
|------------|------------------------------------------------------------------------------------|
| R1         | James M.J. (2000) Modern Food Microbiology, 5th Edition, CBS Publishers            |
| R2         | Barnart, G. J. (1997) Basic Food Microbiology, CBS Publishers                      |
| R3         | William Carroll Frazier and Dennis C Westoff (2017) Food Microbiology, 5th Edition |
| R4         | Bibek Ray (1996) Fundamental Food Microbiology, CRC Press                          |
| R5         | M. R. Adams, M. O. Moss. 2008 Food Microbiology, 2nd Edition                       |

## Online Resources

| Sl.No | WebsiteLink                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://faculty.uobasrah.edu.iq/uploads/teaching/1648242688.pdf">https://faculty.uobasrah.edu.iq/uploads/teaching/1648242688.pdf</a>     |
| 2     | <a href="https://www.britannica.com/science/growth-curve">https://www.britannica.com/science/growth-curve</a>                                     |
| 3     | <a href="https://www.cdc.gov/foodsafety/foodborne-germs.html">https://www.cdc.gov/foodsafety/foodborne-germs.html</a>                             |
| 4     | <a href="https://ispe.org/initiatives/regulatory-resources/gmp/what-is-gmp">https://ispe.org/initiatives/regulatory-resources/gmp/what-is-gmp</a> |